

CACERES, Spain

WMO: 082610

Lat: 39.4714N

Lon: 6.3389W

Elev: 391

StdP: 96.72

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	1.4	-5.6	2.5	6.5	-3.9	2.8	6.5	9.2	10.7	8.3	10.6	1.1	230	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	37.6	19.6	36.2	19.3	34.6	18.9	21.2	33.9	20.4	33.0	19.7	31.9	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	13.0	23.6	16.6	12.4	22.6	15.9	11.8	22.0	62.9	34.0	60.1	33.0	57.8	32.0	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	6.3	-2.4	40.4	1.1	1.6	-3.2	41.6	-3.8	42.5	-4.4	43.4	-5.2	44.6		
			-3.4	23.4	1.2	2.0	-4.2	24.8	-5.0	26.0	-5.6	27.2	-6.5	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	8.2	9.5	12.3	14.4	18.2	23.8	26.5	26.3	22.3	17.8	12.0	9.0
	DBStd	7.20	2.67	2.64	3.01	3.37	3.73	3.98	3.21	3.00	3.23	2.97	2.76	2.69
	HDD10.0	185	69	38	13	3	0	0	0	0	0	11	52	
	HDD18.3	1458	316	247	188	125	49	4	0	4	44	189	291	
	CDD10.0	2643	12	24	83	136	256	413	511	505	369	243	72	20
	CDD18.3	873	0	0	0	8	47	168	252	247	123	28	1	0
	CDH23.3	10381	0	0	6	99	510	1996	3252	3107	1204	208	0	0
	CDH26.7	5203	0	0	0	12	164	997	1802	1676	497	54	0	0
Wind	WSAvg	2.8	2.6	2.9	3.1	3.1	2.9	2.9	2.9	2.7	2.6	2.5	2.7	2.4
Precipitation	PrecAvg	545	59	50	47	50	45	15	4	5	32	88	79	70
	PrecMax	963	244	167	219	148	107	59	36	26	116	220	355	200
	PrecMin	320	0	1	1	3	0	0	0	0	1	7	1	3
	PrecStd	155	52	40	47	34	32	15	7	8	25	52	74	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	18.8	24.0	28.2	33.0	38.1	39.3	39.5	36.7	31.0	22.0	16.9
		MCWB	11.6	12.1	13.6	16.7	17.6	20.2	19.8	20.2	18.4	16.3	15.5	12.9
	2%	DB	14.6	16.9	22.0	26.2	30.5	35.9	37.7	37.4	33.5	27.9	19.6	15.4
		MCWB	11.5	11.5	13.1	15.3	17.0	19.7	19.4	19.7	18.8	15.7	14.3	12.8
	5%	DB	13.4	15.4	20.3	23.9	28.2	34.1	36.3	35.8	31.3	25.4	18.0	14.3
		MCWB	10.9	10.9	12.7	14.7	16.2	19.0	19.1	19.4	17.7	16.1	13.7	12.0
	10%	DB	12.4	14.0	18.4	21.3	26.2	32.3	34.7	34.2	29.4	23.3	16.5	13.1
		MCWB	10.2	10.6	12.0	13.4	15.7	18.5	18.8	18.9	17.3	15.5	13.3	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	14.0	15.2	18.2	19.6	22.0	22.0	22.7	20.9	19.1	17.1	14.6
		MCDB	15.0	16.3	21.3	26.3	29.4	34.8	34.8	33.8	31.1	25.3	19.3	15.5
	2%	WB	12.4	12.9	14.2	16.4	18.1	20.6	20.9	21.2	19.7	18.1	15.8	13.5
		MCDB	13.7	15.1	20.1	24.3	27.5	33.1	34.6	34.0	30.3	23.0	18.0	14.8
	5%	WB	11.5	11.8	13.3	15.2	17.1	19.8	20.1	20.4	18.9	17.4	14.7	12.4
		MCDB	12.9	14.2	18.6	22.4	26.1	31.9	33.5	32.9	28.5	21.9	17.0	13.9
	10%	WB	10.6	11.0	12.4	14.0	16.2	19.0	19.4	19.6	18.2	16.7	13.8	11.3
		MCDB	12.1	13.7	17.3	20.1	24.3	30.6	32.3	31.8	26.8	21.0	15.9	12.9
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.2	9.9	10.2	11.0	12.5	13.7	13.5	11.6	9.4	7.7	7.2
		MCDBR	8.7	10.2	13.3	14.2	14.5	14.6	15.0	15.1	14.7	13.3	10.2	7.9
		MCWBR	5.9	6.0	6.6	6.4	5.3	4.8	4.6	5.1	5.4	5.6	5.6	5.4
	5% WB	MCDBR	6.3	7.8	11.4	12.6	12.8	13.6	14.1	13.3	12.1	9.1	7.2	6.5
		MCWBR	4.8	5.1	6.2	6.2	5.4	4.8	4.8	5.1	4.7	4.5	4.4	5.1
Clear-Sky Solar Irradiance	taub	0.303	0.318	0.351	0.357	0.366	0.368	0.350	0.363	0.356	0.341	0.316	0.302	
	taud	2.504	2.466	2.388	2.382	2.384	2.395	2.412	2.381	2.423	2.490	2.512	2.513	
	Ebn at Noon	873	906	905	918	912	907	920	900	888	867	849	846	
	Edh at Noon	76	91	109	117	119	118	115	116	104	88	75	71	
All-Sky Solar Radiation	RadAvg	2.23	3.22	4.33	5.50	6.70	7.62	7.90	6.89	5.28	3.55	2.46	1.97	
	RadStd	0.32	0.42	0.52	0.53	0.54	0.39	0.20	0.22	0.32	0.33	0.33	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65

Nomenclature: See separate page